Mathematics I

Week 4: Practice for graded assignment

Q1.1

By dividing a polynomial ‘p(x)’ with another polynomial ‘q(x)’ we get ‘h(x)’ as the quotient and ‘r(x)’ as the remainder.

i.e $p(x) = q(x)\underline{h(x)} + r(x)$

The minimum degree of r(x) can be?

a) 0

b) deg (p(x)) - 1

c) 1

~~d)~~ deg (q(x)) - 1

$-x = -1 \times x^{1}$

$\deg(-x) = 1.$

$0 \rightarrow$ undefined.

$1 = 1 \cdot x^0$

$2 = 2 \cdot x^0$

$3.$

$x+1$

$0 = 0 \cdot x^0$

$= 0 \cdot x^1$

$= 0 \cdot x^2$

$P(x) = \underline{0} \leftarrow$ undefined

Solution:

- The degree of the remainder r(x) should be strictly less than the degree of the polynomial q(x).
- Case 1: When remainder is constant (c) but not 0
  - In this case c can be written as cx^0
  - The degree of constant polynomial is 0
- Case 2: When remainder is constant (c) but c = 0
  - In this case the degree is not defined.
- So from the given option the minimum degree will be 0.

$P(x) = x+1$

$q(x) = x$

$x \,\big)\, x+1 \,\big(\, 1$

$\underline{x}$

1

$x+1 = 1.x + 1$ — Re

0

$$P(x) = q(x)\cdot h(x) + \underline{\underline{r(x)}}.$$

$r(x) = 0 \longleftarrow$ ~~undefined~~

$r(x) = -1 \longleftarrow 0$

$r(x) = x+1 \longleftarrow 1.$

$\boxed{-1 \neq -1\cdot x^1}$

$\underline{\underline{-1 = -1\cdot x^0}}$

$\underline{\underline{0, 1, 2, 3, \ldots}}$

$\underline{\underline{0}}$

$0 = 0\cdot x^0$

$= 0\cdot x^1$

$= 0\cdot x^2$

$= 0\cdot x^3$

not defined.

Q1.2

By dividing a polynomial ‘p(x)’ with another polynomial ‘q(x)’ we get ‘h(x)’ as the quotient and ‘r(x)’ as the remainder.

$$p(x) = q(x)h(x) + r(x)$$

The maximum degree of r(x) can be?

a) deg p(x)

b) deg (p(x)) - 1

c) deg q(x)

d) deg (q(x)) - 1

$q(x) = c$

$$\deg(r(x)) < \deg(q(x)).$$

$$\hookrightarrow \deg(r(x)) = \deg(q(x)) - 1.$$

Solution:

- The degree of the remainder r(x) should be strictly less than the degree of the polynomial q(x).
- So the maximum degree of r(x) is deg (q(x))-1.

$q(x) = 2$ $\qquad$ $\underline{\underline{r(x) = 0}}$

$$P(x) = 4x^2+6x-12 = \underset{q(x)}{\underset{\uparrow}{2}}\,\underset{h(x).}{\underset{\uparrow}{(2x^2+3x-6)}} + \underset{\underline{\underline{r(x)}}}{\underset{\uparrow}{\underline{\underline{0}}}}$$

$P(x) = 4x^2+3.$ $\qquad$ $q(x) = 4.$

$$4x^2+3 = 4(x^2+3/4) + 0$$

$p(x) = x^{14} + x^9 \qquad q(x) = x^3$

$$x^{14} + x^9 = x^3(\underbrace{x^{11} + x^6}_{h(x)}) + \underline{\underline{0}} \leftarrow r(x)$$

$$\begin{array}{r|l|l} x^3 & x^{14} + x^9 & x^{11} + x^6 \\ & \underline{x^{14}} & \\ & x^9 & \\ & \underline{x^9} & \\ & 0 & \end{array}$$

$p(x) = x^2$

$q(x) = x^2 + 4$

$p(x) = x^4 + 7 \qquad q(x) = x^2 + 1$

Q2

Consider a polynomial function

$$p(x) = -\frac{1}{100000}(-x-10)^2 (x-8)(x-6)(x-4)(x-2)\,x^2\,(x+9)$$

Hint: Draw the graph of p(x)

Which of the following options is/are correct?

a) The number of turning points of p(x) are 9.

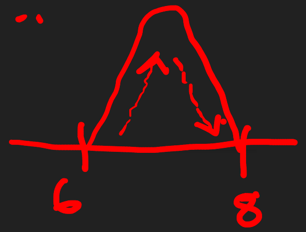
b) p(x) is increasing function when x ϵ (6, 8)

~~c)~~ p(x) is odd function

d) p(x) is one to one function when x ϵ [8, ∞)

$f(-x) = -f(x)$ $f(x) = \sin x$

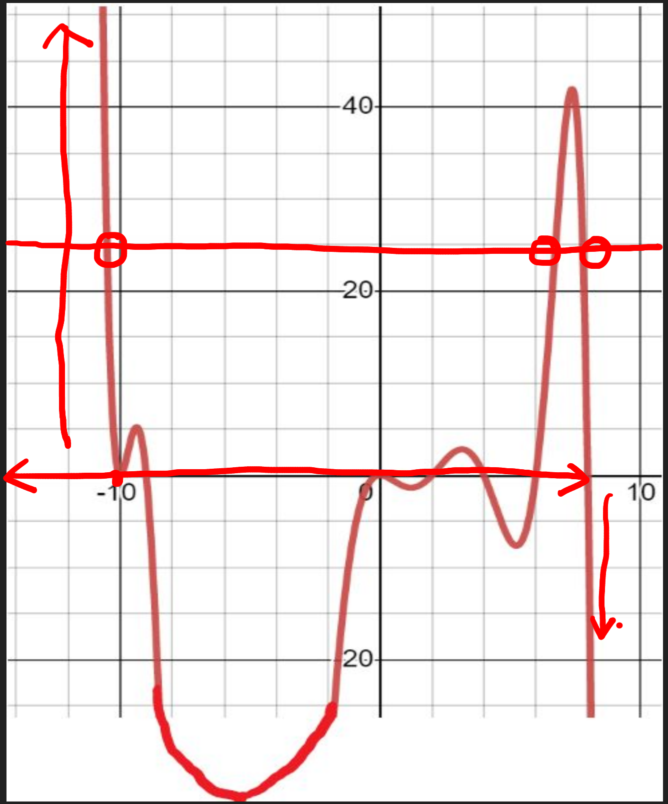
$f(x) = x^n$ + - - - - - -



Solution:

- Step1: Draw a rough sketch of p(x)
  - Rough sketch of p(x) is shown

R



- Step2: Analysis from graph as per given options
  - The number of turning points of p(x) are 8.

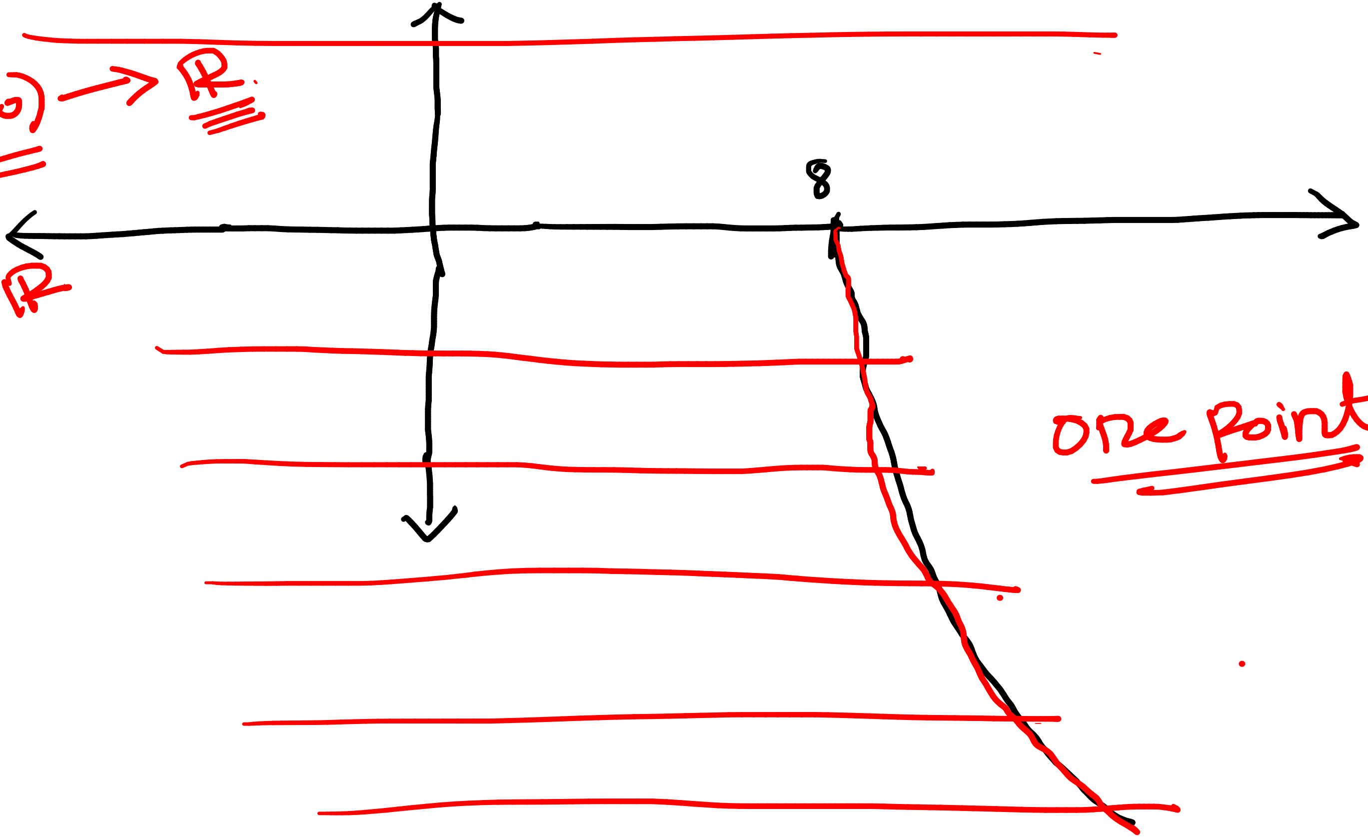
- p(x) is increasing/decreasing function when x ϵ (6, 8)

- p(x) is neither odd nor even function

- p(x) is one to one function when x ϵ $\boxed{[8, \infty)}$

$P(x): [8, \infty) \to \mathbb{R}$

$P(x): \mathbb{R} \to \mathbb{R}$



one point

Q3

Given that m, n and p are the roots of the polynomial $x^3 + 5x - 10$ and sum of the roots is 0.

Find the value of $m^3 + n^3 + p^3$?

$f(x)$: $\boxed{m+n+p=0}$

$f(m)=0 \Rightarrow m^3+5m-10=0$ — ①

$f(n)=0 \Rightarrow n^3+5n-10=0$ — ②

$f(p)=0 \Rightarrow p^3+5p-10=0$ — ③

$f(m)+f(n)+f(p)=0 \Rightarrow m^3+n^3+p^3+5\underbrace{(m+n+p)}_{=0}-30=0$

$\Rightarrow m^3+n^3+p^3-30=0$

$\Rightarrow \boxed{m^3+n^3+p^3=30}$

$x^3+5x-10$

$=(x-p/q)(x^2+bx+c)$

Two complex roots

Solution:

- Given that m, n and p are the roots of $f(x) = x^3 + 5x - 10$, then
  - $f(m) = m^3 + 5m - 10 = 0$
  - $f(n) = n^3 + 5n - 10 = 0$
  - $f(p) = p^3 + 5p - 10 = 0$
  - Then, $f(m) + f(n) + f(p) = 0$
    - $m^3 + n^3 + p^3 + 5(m + n + p) - 30 = 0$ (Given: $m + n + p = 0$)
    - $\rightarrow m^3 + n^3 + p^3 + 5(0) - 30 = 0$
    - $\rightarrow m^3 + n^3 + p^3 = 30$

Q4

The Ministry of Road Transport and Highways wants to connect three aspirational districts with two roads r_1 and r_2. Two roads are connected if they intersect. The shape of the two roads r_1 and r_2 follows polynomial curve $f(x) = (x-11)^2(x-13)^2$ and $g(x) = -(x-11)(x-13)$ respectively.

Hint:

The coordinates of the aspirational districts are at the intersection points of road r_1 and r_2, i.e., when f(x) = g(x)

What will be the *x*-coordinate of the third aspirational district, if the first two are at *x*-intercepts of *f(x)* and *g(x)*.

$1\cancel{1, 13}$ $\boxed{(11,0) \text{ \& } (13,0)}$

$f(x) = g(x) \Rightarrow (x-11)^2(x-13)^2 = -(x-11)(x-13)$

$\Rightarrow (x-11)(x-13)\left[(x-11)(x-13)+1\right] = 0$

$-1 \times -3 + 1 = 4$ ← $(x-11)(x-13)+1=0$ $(12, 0)$

Solution:

$\Rightarrow (x-12)^2 = 0 \Rightarrow \boxed{x=12}$

- Step 1: Two roads are connected when f(x) = g(x)
  - This will give us the x-coordinate of all three aspirational district.
  - $(x-11)^2(x-13)^2 = -(x-11)(x-13)$
  - $(x-11)(x-13)\{(x-11)(x-13)+1\} = 0$ ……(On solving $\{(x-11)(x-13)+1\}$ we get $(x-12)^2$)
  - $\Rightarrow (x-11)(x-13)(x-12)^2 = 0$
  - The above equation will be zero when
    - x = 11 or 12 or 13
- Step 2: Two of the aspirational districts are at x-intercepts of f(x) and g(x)
  - The x-intercepts of $f(x)=(x-11)^2(x-13)^2$ are 11 and 13.
  - The x-intercepts of $g(x) = -(x-11)(x-13)$ are 11 and 13.
  - Thus two of the aspirational districts are at (11, 0) and (13, 0)
  - ⇒ Then, certainly x = 12 is the x-coordinate of the third aspirational district.

Q5

Polynomial fit for the data given in the table recorded by a student is $y = f(x) = x^3+3x^2+n$.

x	-3	-2	-1	0
$y = f(x)$	-2	2	0	-2

"$\underline{\underline{0}}$"

Find the value of $\underline{\underline{\text{'n'}}}$ (till one decimal place), so that SSE (sum squared error) will be minimum?

$$\sum_{i=1}^{4} (y_i - f(x_i))^2.$$

$$4(n+2)^2.$$

$$= 4(n^2+4+4n)$$

Solution:

$\sum_{i=1}^{4}(y_i - f(x_i))^2.$

- Step1:
  - SSE formula
    $\sum_{i=1}^{n} (y_i - (f(x_i)))^2$
  - Calculate SSE as shown in below figure

x	y	$[y-(x^3+3x^2+n)]^2$
-3	-2	$[-2-((-3)^3+3(-3)^2+n)]^2 = \underline{(-2-n)^2}$ $(n+2)^2$
-2	2	$(-2-n)^2 =$
-1	0	$(-2-n)^2 = (n+2)^2.$
0	-2	$(-2-n)^2 =$

$\underline{SSE} = \underline{4(-2-n)^2} = 4(n^2+4n+4)$ $4(n^2+4n+4)$

$= \boxed{4n^2+16n+16}$ $4n^2+$

Minimum at vertex $= -\frac{b}{2a} = \frac{-16}{2\times 4} = -2$ $n =$

$\boxed{n=-2}$

min

- SSE (sum squared error) will be minimum when n = -2

$(0-(-1+3+n))^2$

$\Rightarrow (-(2+n))^2$

$= (n+2)^2$

$$\frac{8x^4+\sqrt{8x^4}}{\sqrt{5x^4}} = \frac{8x^4+2\sqrt{2}\,x^2}{\sqrt{5}\,x^2}$$

$$= \frac{8}{\sqrt{5}}x^2 + \frac{2\sqrt{2}}{\sqrt{5}}$$

$$\sqrt{10x^{16}y^{12}z^{8}} = \left(10x^{16}y^{12}z^{8}\right)^{1/2}$$

$$= (10)^{1/2}\,x^{8}\,y^{6}\,z^{4} \longrightarrow 18$$

$$P(x) = (-)(\overset{x}{x-1})(\overset{x}{x-2})(\overset{x^2}{x-3})^2(\overset{x}{x-4})\,\overset{x}{x}\,(\overset{x}{x+1})(\overset{x^2}{x+2})^2$$

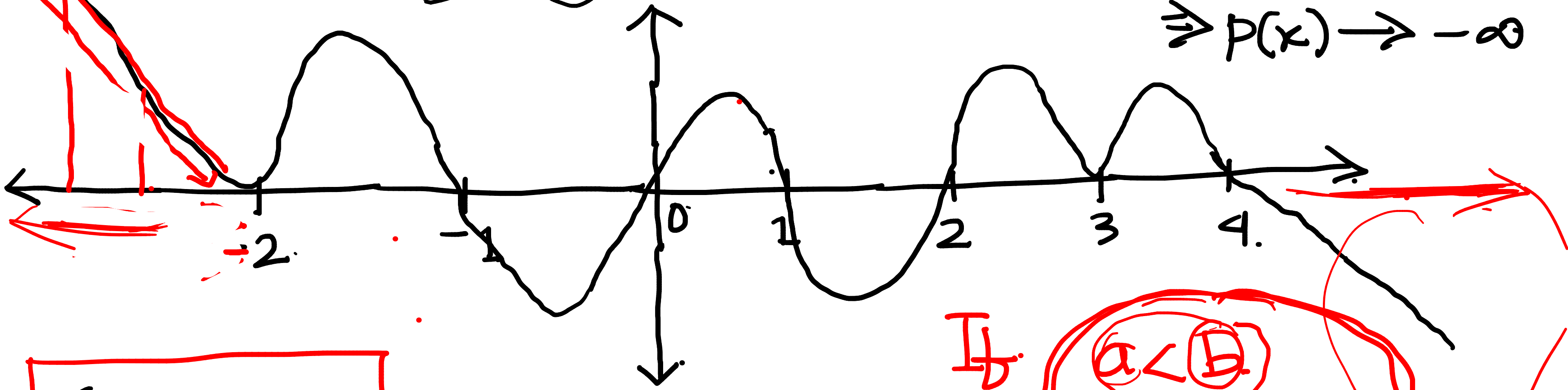
$-2,\ -1,\ 0,\ 1,\ 2,\ 3,\ 4,$

$\downarrow$ 2, 1, 1, 1, 1, 2, 1

$P(x) = \underline{-x^9} + \cdots$

$x \to +\infty \quad -x^9 \to -\infty$

$\Rightarrow P(x) \to -\infty$



$\boxed{(-\infty, -2]}$

$-4 < -3$

?? $f(-4) < f(-3)$

If $a < b$

$f(a) < f(b).$